

APS-02 — your data model

Real-Time Dynamic Pricing & Demand Forecasting

Kognivera Hackathon 2026 · Travel & Tourism · data model v1.1.0-rc1

The problem statement itself, the 24-hour MVP scope and the XR device requirement live in the hackathon application, on your statement's page. This document is the data you have been given to build it with: every table, every field, and what each one is for.

You have **55,033 rows across 13 tables**. 10 of them are the tables this statement is built on; the remaining 3 are reference tables the others point at, included so the database works on its own.

All of it is in the `data/` folder beside this document: as `APS-02.db` (SQLite, indexed, ready to query), as CSV, and as DDL for Postgres and SQLite.

What the data gives you

20,000 pricing events across **13 months** with visible seasonality and lead-time structure, 1,500 sets of price guardrails, and 5,950 rows of computed price history with the driving factors recorded alongside each one.

Watch out for this one

About 1% of room rates are deliberate outliers. A forecast that lets them drag the curve is a forecast that has not been defended — handle them, do not assume they are a mistake in the data.

The tables this statement is built on

TABLE	ROWS	WHAT YOU USE IT FOR
<code>cities</code>	60	The geographic anchor of the whole model. 60 cities; every hotel, POI, package, advisory and weather row hangs off one.
<code>countries</code>	30	ISO country reference. Every city, currency default and calling code resolves here.
<code>currencies</code>	25	carries the true minor-unit exponent so JPY/KWD display correctly even though storage is always DECIMAL(12,2).
<code>flights</code>	4,000	Schedule instances across the search window, so a date search returns results rather than blanks.
<code>hotels</code>	300	Fewer, richer properties. Depth (reviews, room types, media) matters more than catalogue size —.
<code>inventory_calendar</code>	15,030	Entity x date availability with a database-level no-oversell CHECK. Deliberately scarce units so APS-05's load test has something to defend.

TABLE	ROWS	WHAT YOU USE IT FOR
<code>flight_fares</code>	8,002	The bookable unit for flights, and the second <code>entity_type</code> in <code>inventory_calendar</code> .
<code>price_bounds</code>	1,500	The guardrails APS-02 must respect. A dynamic price outside these is a failing demo, not a clever one.
<code>price_history</code>	5,950	Computed price over time with the driving factors alongside — the shape APS-02's explainability requirement has to produce.
<code>pricing_events</code>	20,000	13 months of search / booking / cancellation signal with visible seasonality — otherwise APS-02's forecast is a straight line and the price curve has nothing to explain.

Reference tables, included so the database is valid

You will mostly join through these rather than think about them.

TABLE	ROWS	WHAT IT IS
<code>languages</code>	26	Rule R6: BCP-47 is the only legal way to say 'language' anywhere in the model.
<code>airlines</code>	30	Carrier reference for flight schedule instances.
<code>airports</code>	80	So flights join airport-to-airport rather than city-to-city.

How they fit together

Open `02_DATA_MODEL_DIAGRAM.html` in a browser for the clickable version — it shows these tables and nothing else. Download it first; it will not render inside SharePoint.

Some tables point at "any bookable thing" using an `(entity_type, entity_id)` pair rather than a typed foreign key. That is deliberate: it is what lets one feature refer to a hotel, a flight, a point of interest or a package without a separate join table for each. The legal values of `entity_type` are in `data/enums.json`.

Every field, table by table

Columns marked **PK** are the primary key. **FK** shows what a column points at. Enum columns list their legal values — anything else is rejected by the conformance check.

`currencies`

Reference & geography · 25 rows · IDs start `cur_`

carries the true minor-unit exponent so JPY/KWD display correctly even though storage is always `DECIMAL(12,2)`.

COLUMN	TYPE	CONSTRAINTS	NOTES
currency_id	text	PK	cur_ prefixed.
iso4217	char(3)	UNIQUE · NOT NULL	e.g. INR.
name	text	NOT NULL	
symbol	text	NOT NULL	
minor_unit_exponent	smallint	NOT NULL	0 for JPY/KRW, 2 default, 3 for KWD/BHD.
display_locale	text	NOT NULL	BCP-47 locale used for formatting.
updated_at	timestampz	NOT NULL	

inventory_calendar

Availability & pricing · 15,030 rows · IDs start `inv_`

Entity x date availability with a database-level no-oversell CHECK. Deliberately scarce units so APS-05's load test has something to defend.

COLUMN	TYPE	CONSTRAINTS	NOTES
inventory_id	text	PK	inv_ prefixed.
entity_type	text	NOT NULL · one of <code>room_type</code> , <code>flight_fare</code> , <code>guide</code> , <code>poi</code>	
entity_id	text	NOT NULL	room_type_id or fare_id.
for_date	date	NOT NULL	
total_units	int	NOT NULL	
booked_units	int	NOT NULL	
held_units	int	NOT NULL	
price	decimal(12,2)	NOT NULL	the price for that date.
currency	char(3)	FK → <code>currencies.iso4217</code> · NOT NULL	
min_stay_nights	smallint	NOT NULL	
closed_to_arrival	bool	NOT NULL	
updated_at	timestampz	NOT NULL	

Unique together: (`entity_type`, `entity_id`, `for_date`)

Enforced by the database: `booked_units + held_units <= total_units`

languages

Reference & geography · 26 rows · IDs start `lng_` · reference table

Rule R6: BCP-47 is the only legal way to say 'language' anywhere in the model.

COLUMN	TYPE	CONSTRAINTS	NOTES
language_id	text	PK	lng_ prefixed.
bcp47	text	UNIQUE · NOT NULL	e.g. ta, hi, en-IN — never 'Tamil'.
english_name	text	NOT NULL	
native_name	text	NOT NULL	
script	text	NOT NULL	ISO-15924, e.g. Taml, Deva, Latn.
rtl	bool	NOT NULL	Right-to-left rendering flag.
tts_supported	bool	NOT NULL	Relevant to PS-13 voice output.
updated_at	timestampz	NOT NULL	

price_bounds

Availability & pricing · 1,500 rows · IDs start pbd_

The guardrails APS-02 must respect. A dynamic price outside these is a failing demo, not a clever one.

COLUMN	TYPE	CONSTRAINTS	NOTES
bound_id	text	PK	pbd_ prefixed.
entity_type	text	NOT NULL · one of room_type, flight_fare, guide, poi	
entity_id	text	UNIQUE · NOT NULL	
floor_price	decimal(12,2)	NOT NULL	D3.
ceiling_price	decimal(12,2)	NOT NULL	
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
max_daily_move_pct	decimal(5,2)	NOT NULL	Cap on day-over-day movement.
max_weekly_move_pct	decimal(5,2)	NOT NULL	
rounding_step	decimal(12,2)	NOT NULL	Prices are rounded to this step for display.
override_active	bool	NOT NULL	Manual admin override in force.
updated_at	timestampz	NOT NULL	

price_history

Availability & pricing · 5,950 rows · IDs start phs_

Computed price over time with the driving factors alongside — the shape APS-02's explainability requirement has to produce.

COLUMN	TYPE	CONSTRAINTS	NOTES
history_id	text	PK	phs_ prefixed.
entity_type	text	NOT NULL · one of room_type, flight_fare, guide, poi	
entity_id	text	NOT NULL	
effective_date	date	NOT NULL	
price	decimal(12,2)	NOT NULL	D3.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
baseline_price	decimal(12,2)	NOT NULL	
demand_index	decimal(6,3)	NOT NULL	Forecast output that drove the move.
occupancy_pct	decimal(5,2)	NOT NULL	
lead_time_factor	decimal(6,3)	NOT NULL	
seasonality_factor	decimal(6,3)	NOT NULL	
event_factor	decimal(6,3)	NOT NULL	Uplift from a festival in events_festivals.
competitor_factor	decimal(6,3)	NOT NULL	
bound_clamped	bool	NOT NULL	True where price_bounds clipped the computed price.
explanation	text	NOT NULL	Human-readable factor summary.
computed_at	timestampz	NOT NULL	

Unique together: (entity_type, entity_id, effective_date)

countries

Reference & geography · 30 rows · IDs start cnt_

ISO country reference. Every city, currency default and calling code resolves here.

COLUMN	TYPE	CONSTRAINTS	NOTES
country_id	text	PK	Canonical ID, cnt_ prefixed.
iso2	char(2)	UNIQUE · NOT NULL	ISO-3166-1 alpha-2, e.g. IN.
iso3	char(3)	UNIQUE · NOT NULL	ISO-3166-1 alpha-3, e.g. IND.

COLUMN	TYPE	CONSTRAINTS	NOTES
name	text	NOT NULL	English short name.
default_currency	char(3)	FK → currencies.iso4217 · NOT NULL	ISO-4217 code.
calling_code	text	NOT NULL	E.164 country calling code, e.g. +91.
region	text	NOT NULL	UN sub-region grouping.
updated_at	timestampz	NOT NULL	Rule R4: UTC, ISO-8601 with offset.

airlines

Reference & geography · 30 rows · IDs start arl_ · reference table

Carrier reference for flight schedule instances.

COLUMN	TYPE	CONSTRAINTS	NOTES
airline_id	text	PK	arl_ prefixed.
iata	char(2)	UNIQUE · NOT NULL	Carrier code, e.g. 6E.
name	text	NOT NULL	
alliance	text		star_alliance
country_id	text	FK → countries.country_id · NOT NULL	
low_cost	bool	NOT NULL	Drives fare banding.
updated_at	timestampz	NOT NULL	

cities

Reference & geography · 60 rows · IDs start cty_

The geographic anchor of the whole model. 60 cities; every hotel, POI, package, advisory and weather row hangs off one.

COLUMN	TYPE	CONSTRAINTS	NOTES
city_id	text	PK	cty_ prefixed.
name	text	NOT NULL	City name.
state	text		State / province, nullable for city-states.
country_id	text	FK → countries.country_id · NOT NULL	
country_code	char(2)	NOT NULL	Denormalised ISO2 for convenient joins.

COLUMN	TYPE	CONSTRAINTS	NOTES
lat	decimal(9,6)	NOT NULL	Rule R7: WGS-84, 6dp.
lng	decimal(9,6)	NOT NULL	Rule R7: WGS-84, 6dp.
timezone	text	NOT NULL	IANA zone, e.g. Asia/Kolkata.
region	text	NOT NULL	Domestic region grouping, e.g. South India.
population	int		Approximate, for demand weighting.
season_profile	text	NOT NULL · one of winter, summer, monsoon, post_monsoon, spring, autumn	Dominant season at the peak travel window.
peak_months	text	NOT NULL	Comma-separated month numbers, e.g. 10,11,12.
primary_language	text	FK → languages.bcp47 · NOT NULL	Rule R6: BCP-47 tag.
description	text		One-paragraph orientation blurb, used by PS-13.
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestampz	NOT NULL	

hotels

Supply & catalogue · 300 rows · IDs start `htl_`

Fewer, richer properties. Depth (reviews, room types, media) matters more than catalogue size —.

COLUMN	TYPE	CONSTRAINTS	NOTES
hotel_id	text	PK	htl_ prefixed.
city_id	text	FK → cities.city_id · NOT NULL	
name	text	NOT NULL	~3% near-duplicate names injected deliberately.
property_type	text	NOT NULL · one of hotel, resort, homestay, hostel, apartment, boutique, heritage, guesthouse	
star_rating	smallint	NOT NULL	1–5, CHECK constrained.
guest_score	decimal(2,1)		0.0–10.0; null for a handful of new

COLUMN	TYPE	CONSTRAINTS	NOTES
			properties.
review_count	int	NOT NULL	Denormalised count, must agree with hotel_reviews.
address_line	text	NOT NULL	
lat	decimal(9,6)	NOT NULL	
lng	decimal(9,6)	NOT NULL	
distance_to_centre_km	decimal(6,2)	NOT NULL	PS-02 filter: distance to a landmark.
description	text	NOT NULL	Plausible prose, embeddable.
base_currency	char(3)	FK → currencies.iso4217 · NOT NULL	
checkin_time	text	NOT NULL	Local HH:MM at the property.
checkout_time	text	NOT NULL	
chain_code	text		Null for independents.
has_xr_scene	bool	NOT NULL	PS-05 / APS-07 — does an immersive preview exist.
status	text	NOT NULL · one of active, inactive, archived, draft	
created_at	timestampz	NOT NULL	
updated_at	timestampz	NOT NULL	

pricing_events

Availability & pricing · 20,000 rows · IDs start pev_

13 months of search / booking / cancellation signal with visible seasonality — otherwise APS-02's forecast is a straight line and the price curve has nothing to explain.

COLUMN	TYPE	CONSTRAINTS	NOTES
event_id	text	PK	pev_ prefixed.
entity_type	text	NOT NULL · one of room_type, flight_fare, guide, poi	
entity_id	text	NOT NULL	
city_id	text	FK → cities.city_id · NOT NULL	Denormalised for fast grouping.

COLUMN	TYPE	CONSTRAINTS	NOTES
event_type	text	NOT NULL · one of search, view, booking, cancellation, abandon	
occurred_at	timestamptz	NOT NULL	R4.
for_date	date	NOT NULL	The stay/travel date being searched or booked.
lead_time_days	int	NOT NULL	for_date minus occurred_at date — a key forecast feature.
channel	text	NOT NULL · one of web, mobile_app, partner, call_centre, agent	
party_size	smallint	NOT NULL	
quoted_price	decimal(12,2)		null for pure searches.
currency	char(3)	FK → currencies.iso4217	
converted	bool	NOT NULL	Did this search become a booking.
session_id	text	NOT NULL	Opaque session grouping key.

airports

Reference & geography · 80 rows · IDs start apt_ · reference table

So flights join airport-to-airport rather than city-to-city.

COLUMN	TYPE	CONSTRAINTS	NOTES
airport_id	text	PK	apt_ prefixed.
iata	char(3)	UNIQUE · NOT NULL	IATA code.
icao	char(4)		ICAO code.
name	text	NOT NULL	
city_id	text	FK → cities.city_id · NOT NULL	
lat	decimal(9,6)	NOT NULL	
lng	decimal(9,6)	NOT NULL	
timezone	text	NOT NULL	IANA.
is_international	bool	NOT NULL	
updated_at	timestamptz	NOT NULL	

flights

Supply & catalogue · 4,000 rows · IDs start flt_

Schedule instances across the search window, so a date search returns results rather than blanks.

COLUMN	TYPE	CONSTRAINTS	NOTES
flight_id	text	PK	flt_ prefixed.
airline_id	text	FK → airlines.airline_id · NOT NULL	
flight_number	text	NOT NULL	e.g. 6E-2043.
origin_airport_id	text	FK → airports.airport_id · NOT NULL	
dest_airport_id	text	FK → airports.airport_id · NOT NULL	
departs_at	timestampz	NOT NULL	Rule R4: carries offset.
arrives_at	timestampz	NOT NULL	
duration_minutes	int	NOT NULL	
stops	smallint	NOT NULL	
aircraft_type	text		
cabin_classes	text	NOT NULL	Comma-separated cabin_class values available.
carbon_kg	decimal(8,3)	NOT NULL	Per-seat estimate — APS-09 reads this.
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestampz	NOT NULL	

flight_fares

Supply & catalogue · 8,002 rows · IDs start far_

The bookable unit for flights, and the second entity_type in inventory_calendar.

COLUMN	TYPE	CONSTRAINTS	NOTES
fare_id	text	PK	far_ prefixed.
flight_id	text	FK → flights.flight_id · NOT NULL	
cabin_class	text	NOT NULL · one of economy, premium_economy, business, first	
fare_class	text	NOT NULL · one of saver, flex, standard, corporate, promo	
base_fare	decimal(12,2)	NOT NULL	D3.

COLUMN	TYPE	CONSTRAINTS	NOTES
taxes	decimal(12,2)	NOT NULL	
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
baggage_kg	smallint	NOT NULL	
cabin_baggage_kg	smallint	NOT NULL	
changeable	bool	NOT NULL	
change_fee	decimal(12,2)		
refundable	bool	NOT NULL	
seats_total	int	NOT NULL	
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestampz	NOT NULL	

The rules that apply to these fields

#	RULE
R1	Additive only. Add columns, tables and stores freely. Never rename, drop or repurpose a field that came with the data.
R2	IDs are opaque prefixed strings — <code>htl_a91f3c</code> . Never integers, never parsed for meaning.
R3	Money is a pair: a 2-place decimal plus an ISO-4217 currency code. Never a float.
R4	Time is ISO-8601 with an offset. <code>_at</code> fields carry an offset; <code>_date</code> fields have no zone.
R5	Enums are lowercase snake_case and the legal values are in <code>data/enums.json</code> .
R6	Language is a BCP-47 tag — <code>ta</code> , not "Tamil".
R7	Geography is WGS-84 to 6 decimal places, <code>lat</code> and <code>lng</code> together or not at all.
R8	Nothing is hard-deleted. Rows carry <code>status</code> and <code>updated_at</code> .

Add whatever you like beside these fields — new columns, new tables, your own vector store, your own services. That is the point of R1. What you must not do is rename or re-key the fields that came with the data, because that is what would stop sixteen independent builds being put together afterwards.

`data/WORKING_WITH_THE_DATA.md` has the loading instructions, including how to read money without corrupting it. `tools/validate_conformance.py` tells you in thirty seconds whether you are still conformant.